Fiscal Year:	FY 2020	**Task Last Updated:**	FY 08/03/2021
PI Name:	Simpson, Richard Ph.D.		
Project Title:	The Impact of Modeled Microgravity and Prior Radiation Exposure on Cytomegalovirus Reactivation and Host Immune Evasion		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Immune**:Risk of In Mission Impacts, Adverse Health Events or Long-Term Health Impacts due to Altered Immune Response		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rjsimpson@email.arizona.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	713-397-0121
Organization Name:	University of Arizona		
PI Address 1:	College of Agriculture and Life Sciences; College of Medicine		
PI Address 2:	1177 E. Fourth Street, Room 308, Shantz Building		
PI Web Page:			
City:	Tucson	**State:**	AZ
Zip Code:	85721-0001	**Congressional District:**	3
Comments:	NOTE: Formerly at University of Houston until September 2017 move to University of Arizona.		
Project Type:	Ground	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-Crew Health (FLAGSHIP & NSBRI)
Start Date:	10/21/2015	**End Date:**	04/30/2020
No. of Post Docs:	3	**No. of PhD Degrees:**	5
No. of PhD Candidates:	6	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
Contact Email:	Peter.norsk@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 4/30/2020 per NSSC information (Ed., 1/16/2020) NOTE: End date changed to 9/30/2019 per NSSC information (Ed., 3/4/19) NOTE: End date changed to 10/20/2018 per NSSC information (Ed., 3/5/18) NOTE: Element change to Human Health Countermeasures; previously Space Human Factors & Habitability (Ed., 1/18/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Crucian, Brian Ph.D. (NASA Johnson Space Center) Laughlin, Mitzi Ph.D. (University of Houston) Markofski, Melissa Ph.D. (University of Houston) Mehta, Satish Ph.D. (NASA Johnson Space Center) Pierson, Duane Ph.D. (NASA Johnson Space Center) Rezvani, Katayoun M.D., Ph.D. (The University of Texas MD Anderson Cancer Cente)		

Grant/Contract No.:	NNX16AB29G
Performance Goal No.:	
Performance Goal Text:	
Task Description:	Cytomegalovirus (CMV), which infects 50-80% of the US population, and CMV DNA is frequently found in the body fluids of astronauts during space travel indicating that the spaceflight environment can initiate active CMV infections. The effects of microgravity on CMV replication and its potential to evade the host immune system are unknown. In this study, we will utilize the rotating wall vessel (RWV) cell culture system to simulate microgravity on Earth allowing us to study the virulent and infectivity properties of CMV and how it interacts with host immune cells in the absence of gravity. The study hypothesizes that modeled microgravity will evoke changes in cellular gene expression that will lead to CMV reactivation and increase virus replication and infectivity. The researchers believe that modeled microgravity will impair the anti-CMV responses of host T-cells and NK-cells that will allow viral persistence and replication to occur. During testing both traditional and cutting-edge laboratory techniques that are typically used to assess virulence and immune cell function in the transplant patient setting allowing determination if CMV infection in the microgravity environment is likely to increase the risk of an adverse health event during exploration class spaceflight missions. Specific Aims: To determine the impact of simulated microgravity on CMV replication and its potential to evade the host immune system a) To determine the effects of modeled microgravity on CMV replication and the expression of genes associated with latency and lytic activity. b) To determine the impact of modeled microgravity on the anti-CMV activity of host T-cells and NK-cells. This study will provide preliminary data to determine if CMV virus displays increased infectivity when exposed to a low shear modeled microgravity environment. Data will be evaluated and added to the evidence base capturing the responses to microorganisms grown in rotating wall vessels.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The reactivation of latent herpesviruses poses a significant risk to the health and safety of the crew during exploration class spaceflight missions. Of particular concern is cytomegalovirus (CMV), which infects 50-80% of the US population and is known to cause significant morbidity and mortality in immunocompromised patients. CMV DNA is frequently found in the body fluids of astronauts during space travel indicating that the spaceflight environment can initiate active CMV infections. As spaceflight also causes huge alterations in the normal functioning of the immune system, CMV reactivation in a potentially immunosuppressed crewmember could have disastrous consequences during long-duration exploration class missions. The effects of microgravity on CMV replication and its potential to evade the host immune system are unknown. In this study, we utilized the rotating wall vessel (RWV) cell culture system to simulate microgravity on Earth allowing us to study the virulent and infectivity properties of CMV and how it interacts with host immune cells in the absence of gravity. We hypothesized that modeled microgravity will evoke changes in cellular gene expression that will underpin CMV reactivation and increase virus replication and infectivity. We further postulated that modeled microgravity would impair the anti-CMV responses of host T-cells and NK-cells that will allow viral persistence and replication to occur. We concluded that SMG exposure may promote short-term transcription of lytic CMV genes and impair certain aspects of cellular immunity that are pertinent to recognize and control the spread of CMV infections and prevent tumor growth. Microgravity should therefore be considered a potential contributing factor to CMV reactivation and immune dysregulation in long duration exploration mission crew. Studies like these are required so that effective countermeasures can be developed that would allow CMV-infected crewmembers to embark on such missions. Identifying viral reactivation countermeasures to safeguard space travellers could also have important implications for people on Earth, particularly immunocompromised individuals (e.g., transplant recipients) who are at high risk of morbidity and mortality due to viral reactivation.
	There were two major arms of this project. The first arm involved work with live CMV virus to determine the effects of simulated microgravity (SMG) on CMV infectivity, replication, and the expression of genes associated with latency and lytic activity (Aim 1&2). This work was predominantly completed at NASA-Johnson Space Center (JSC) and formed a major part of Bridgette Rooney's PhD dissertation that was supervised by Dr. Simpson (University of Houston/University of Arizona), Dr. Mehta (NASA-JSC), Dr. Melissa Markofski (University of Houston), Dr. Emily LaVoy (University of Houston), and Dr. Ritesh Tandon (University of Mississippi Medical Center). The second arm was focused on the effects of SMG on the anti-tumor and anti-viral activities of human NK-cells, gamma-delta T-cells, and CMV-specific T-cells (Aim 3). This work was completed at the University of Houston and the University of Arizona and formed a major part of Preteesh Leo Mylabathula's PhD dissertation that was supervised by Dr. Simpson (University of Houston/University of Arizona), Dr. Brian Crucian (NASA-JSC), Dr. Emmanuel Katsanis (The University of Arizona), Dr. Yoonjung Park (University of Houston), and Dr. Melissa Markofski (University of Houston). One peer-reviewed scientific manuscript has been published to date that was directly related to the aims of this project (Ed. note 8/3/21: there are other publications from this grant as listed in the Bibliography section below and the Cumulative Bibliography): Mylabathula P, Li L, Bigley A, Markofski M, Crucian B, Mehta S, Pierson D, Laughlin M, Rezvani K, Simpson RJ. Simulated microgravity disarms human NK-cells and inhibits anti-tumor cytotoxicity in vitro. Acta Astronautica. 2020 Sep;174:32-40. Chapter 4 of Dr. Bridgette Rooney's dissertation contains the methods and data that is relevant to Aims 1 & 2 of this project. The peer-reviewed article published in Acta Astronautica and Chapter 4 of Dr. Mylabathula's PhD dissertation contains the methods and data that are relevant to Aim 3 of this project. Main Findings of the Project Specific Aim 1

Task Progress:	The overarching aim was to investigate the effects of SMG on the capacity for a live human CMV to infect human Kasumi-3 myeloid progenitor cells in vitro and to determine the influence of SMG on viral load. We hypothesized that exposing Kasumi-3 cells to SMG would enhance their susceptibility to CMV infection and increase viral replication over multiple days in culture. The most notable findings were: 1. Kasumi-cells exposed to SMG become more resistant to CMV infection. Cells not exposed to SMG had a 2.5x greater viral load. SMG also increased the time before infectious viral particles could be detected by approximately 24 h. 2. SMG promoted a higher expression of lytic transcripts (e.g., UL122) among CMV infected Kasumi-3 cells without affecting latent transcripts (e.g., US28 and UL138). Specific Aim 2 The overarching aim was to determine the effects of SMG on CMV's virulence (i.e., capacity to generate disease) and continued capacity to infect (bind to and enter) fibroblasts. Our hypothesis was that SMG will increase CMV virulence causing greater plaque forming units (PFUs) after SMG in culture as well as promoting retention of its ability to infect human fibroblasts. While this work was not completed to the extent desired due to expiration of the grant and the COVID-19 pandemic interfering with the laboratory work, the following preliminary findings were made: 1. SMG did not appear to affect changes in viral titers (PFUs) over 7-days in culture when compared to 1G controls. 2. SMG did not appear to alter the timing or capacity of the virus to infect human fibroblasts. Specific Aim 3 The overarching aim was to determine the impact of short-term SMG exposure on the ability of human lymphocytes (NK-cells, virus-specific T-cells, and gamma-delta-T cells) to recognize and kill tumor target cells and CMV antigens. We hypothesized that SMG would impair the anti-tumor activity of NK-cells and gamma-delta-T cells as well as the potential for T-cells to expand when presented with CMV antigens and for them to recognize and kill CMV-infected targets. The most notable findings were: 1. Short-term (12 h) exposure to simulated microgravity (SMG) impaired the ability of human blood NK-cells to kill several tumor target cells in vitro when returned to the 1G environment. The reduced ability to kill tumor target cells in vitro was accompanied by an impaired ability to secrete cytolytic granules and effector cytokines. NK-cell surface phenotypes and ability to form conjugates (immunological synapses) were not impaired by SMG. SMG does not appear to alter the susceptibility of K562 tumor cells to NK-cell mediated killing. These data are included in the Acta Astronautica publication. 2. Short-term exposure to SMG impaired the ability of human lymphocytes to recognize CMV antigens and expand into CMV-specific T-cells. However, pre-existing CMV-specific T-cells retained their ability to recognize CMV antigens and secrete effector cytokines even after being exposed to short-term SMG. Similarly, the ability of pre-existing CMV-specific T-cells to recognize and kill CMV peptide pulsed autologous target cells was not impaired by SMG. 3. Short-term exposure to SMG impaired the in vitro expansion of human gamma-delta-T cells. The expression of many activating and inhibitory receptors on the expanded gamma-delta-T cells were not affected by SMG, with the exception of the inhibitory receptor, CD158b, which was found to be downregulated. This perhaps explained why expanded gamma-delta-T cells subsequently exposed to SMG were actually more capable of killing certain tumor target cells in vitro. Despite inherent limitations with our in vitro models in assessing CMV infectivity and anti-viral immunological control, our overarching conclusion is that SMG exposure may promote short-term transcription of lytic CMV genes and impair certain aspects of cellular immunity that are pertinent to recognize and control the spread of CMV infections and prevent tumor growth. Microgravity should be considered as a potential contributing factor to CMV reactivation and immune dysregulation in long duration exploration mission crew.

Bibliography Type:	Description: (Last Updated: 01/12/2026)
Articles in Peer-reviewed Journals	Bigley AB, Agha NH, Baker FL, Spielmann G, Kunz HE, Mylabathula PL, Rooney B, Laughlin MS, Pierson DL, Mehta SK, Crucian BE, Simpson RJ. "N-cell function is impaired during long-duration spaceflight." J Appl Physiol (1985). 2019 Apr 1;126(4):842-53. https://doi.org/10.1152/japplphysiol.00761.2018 ; PMID: 30382809 , Apr-2019
Articles in Peer-reviewed Journals	Agha NH, Baker FL, Kunz HE, Spielmann G, Mylabathula PL, Rooney BV, Mehta SK, Pierson DL, Laughlin MS, Markofski MM, Crucian BE, Simpson RJ. "Salivary antimicrobial proteins and stress biomarkers are elevated during a 6-month mission to the International Space Station." J Appl Physiol (1985). 2020 Feb 1;128(2):264-75. Epub 2019 Nov 21. https://doi.org/10.1152/japplphysiol.00560.2019 ; PMID: 31751178; PMCID: PMC7864226 , Feb-2020
Articles in Peer-reviewed Journals	Agha NH, Mehta SK, Rooney BV, Laughlin MS, Markofski MM, Pierson DL, Katsanis E, Crucian BE, Simpson RJ. "Exercise as a countermeasure for latent viral reactivation during long duration space flight." FASEB J. 2020 Feb;34(2):2869-81. https://doi.org/10.1096/fj.201902327R ; PMID: 31908052 , Feb-2020
Articles in Peer-reviewed Journals	Baker FL, Bigley AB, Agha NH, Pedlar CR, O'Connor DP, Bond RA, Bollard CM, Katsanis E, Simpson RJ. "Systemic ß-adrenergic receptor activation augments the ex vivo expansion and anti-tumor activity of Vgamma9Vdelta2 T-cells." Front Immunol. 2020 Jan 24;10:3082. https://doi.org/10.3389/fimmu.2019.03082 ; PMID: 32038628; PMCID: PMC6993603 , Jan-2020
Articles in Peer-reviewed Journals	Kunz HE, Agha NH, Hussain M, LaVoy EC, Smith KA, Mylabathula P, Diak D, Baker FL, O'Connor DP, Bond RA, Katsanis E, Bollard CM, Simpson RJ. "The effects of ß1 and ß1+2 adrenergic receptor blockade on the exercise-induced mobilization and ex vivo expansion of virus-specific T cells: Implications for cellular therapy and the anti-viral immune effects of exercise." Cell Stress Chaperones. 2020 Nov;25(6):993-1012. https://doi.org/10.1007/s12192-020-01136-7 ; PubMed PMID: 32779001; PubMed Central PMCID: PMC7591642 , Nov-2020

Articles in Peer-reviewed Journals	Mylabathula PL, Diak DM, Baker FL, Niemiro GM, Markofski MM, Crucian BE, Katsanis E, Simpson RJ. "IL-2 and zoledronic acid therapy restores the in vivo anti-leukemic activity of human lymphocytes pre-exposed to simulated microgravity." Front Biosci (Landmark Ed). 2022 Jul 8;27(7):215. Review. https://doi.org/10.31083/j.fbl2707215 ; PMID: 35866402 , Jul-2022
Articles in Peer-reviewed Journals	Mylabathula PL, Li L, Bigley AB, Markofski MM, Crucian BE, Mehta SK, Pierson DL, Laughlin MS, Rezvani K, Simpson RJ. "Simulated microgravity disarms human NK-cells and inhibits anti-tumor cytotoxicity in vitro." Acta Astronaut. 2020 Sep;174:32-40. https://doi.org/10.1016/j.actaastro.2020.03.023 , Sep-2020
Dissertations and Theses	Mylabathula PL. "The Effect of Simulated Microgravity on Anti-CMV and Anti-Leukemia Immune Responses in vitro and in vivo." PhD Dissertation, University of Houston, July 2020. , Jul-2020
Dissertations and Theses	Rooney BV. "Latent Herpes Virus Reactivation iIn Space and the Effects of Simulated Microgravity on CMV Infection." PhD Dissertation, University of Houston, August 2020. , Aug-2020